THE CAP WITH A FIRST CHERN CLASS OF AN INVERTIBLE SHEAF IS THE CLASS OF THE ZERO SCHEME OF A SECTION, UNDER DIMENSION AND GENERIC INJECTIVITY HYPOTHESES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The cap with a first Chern class computed on one integral closed subscheme through a regular section. Same chapter chow.tex, label lemma-compute-c1 (invoked at proof line 4464). Applied to the restriction $s_i = s|_{Y_i}$, it identifies $[Z(s_i)]_k$ with the representative of $c_1(\mathcal{L}|_{Y_i})$ on that component.
- The regrouping of the contributions of the irreducible components of the zero scheme. Same chapter chow.tex, label lemma-prepare-geometric-cap (invoked at proof line 4470). It turns the weighted sum $\sum n_i [Z(s_i)]_k$ into $[Z(s)]_k$, completing the identification.

Source reference. Original lemma 02SK: sources/stacks-chow/pinned-original-source.tex, source lines 4198–4198. Navigation pointer for \ref{lemma-compute-c1} used inside the counted span (line 4464); the argument that consumes it is retained in the span.

Source reference. Original lemma 0EPI: sources/stacks-chow/pinned-original-source.tex, source lines 4358–4358. Navigation pointer for \ref{lemma-prepare-geometric-cap} used inside the counted span (line 4470); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 4433); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $Y \subset X$ be a closed subscheme. Let $s \in \Gamma(Y, \mathcal{L}|_Y)$. Assume*

(1) $\dim_\delta(Y) \leq k + 1$,
(2) $\dim_\delta(Z(s)) \leq k$, *and*
(3) *for every generic point ξ of an irreducible component of $Z(s)$ of δ-dimension k the multiplication by s induces an injection $\mathcal{O}_{Y,\xi} \to (\mathcal{L}|_Y)_\xi$*[1].

Then

$$c_1(\mathcal{L}) \cap [Y]_{k+1} = [Z(s)]_k$$

in $\mathrm{CH}_k(X)$.

Proof. Write

$$[Y]_{k+1} = \sum n_i [Y_i]$$

where $Y_i \subset Y$ are the irreducible components of Y of δ-dimension $k+1$ and $n_i > 0$. By assumption the restriction $s_i = s|_{Y_i} \in \Gamma(Y_i, \mathcal{L}|_{Y_i})$ is not zero, and hence is a regular section. By Lemma 02SK we see that $[Z(s_i)]_k$ represents $c_1(\mathcal{L}|_{Y_i})$. Hence by definition

$$c_1(\mathcal{L}) \cap [Y]_{k+1} = \sum n_i [Z(s_i)]_k$$

Thus the result follows from Lemma 0EPI. □

[1]For example, this holds if s is a regular section of $\mathcal{L}|_Y$.